



Disaster Impact Assessment (DIA) Challenge

Multimodal Deep Learning for Building Damage Classification

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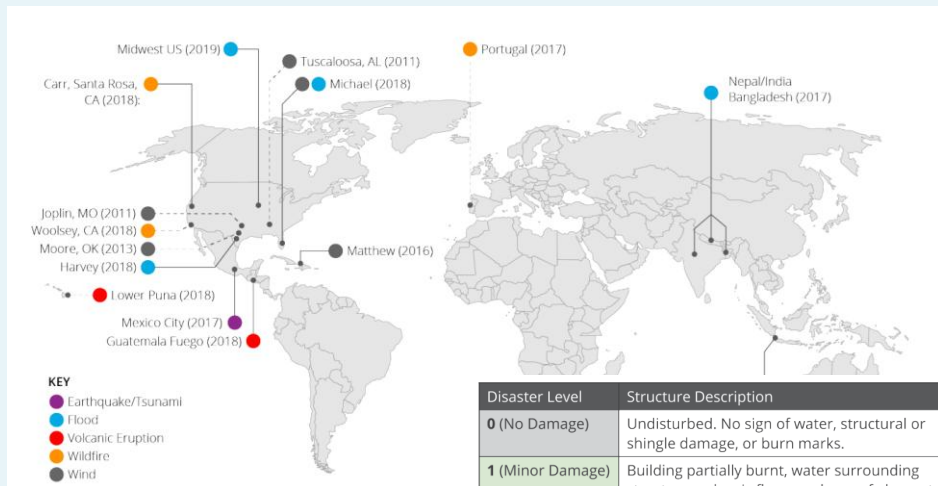
Disaster Impact Assessment (DIA) Challenge

- Multimodal models for building damage classification
- Deep learning and explainable AI (XAI) to assess natural disaster impacts
- Input data: High-resolution pre- and post-disaster satellite imagery, ERA5 climate variables
- Target: 0 – No damage, 1 – Minor damage, 2 – Major damage, 3 – Destroyed
- Predict and understand how environmental conditions contribute to damage from events like floods, wildfires, hurricanes, and volcanic eruptions
- Temporal modeling to capture changes before, during, and after events
- Computational environment based on Python and PyTorch

xBD Dataset

Adapted from [Gupta et al., 2019 / xView2]

High-resolution satellite imagery (<0.8 m GSD) + ERA5 climate variables covering 19 natural disasters worldwide — earthquakes, floods, wildfires, hurricanes, and tsunamis.



Disaster Level	Structure Description
0 (No Damage)	Undisturbed. No sign of water, structural or shingle damage, or burn marks.
1 (Minor Damage)	Building partially burnt, water surrounding structure, volcanic flow nearby, roof elements missing, or visible cracks.
2 (Major Damage)	Partial wall or roof collapse, encroaching volcanic flow, or surrounded by water/mud.
3 (Destroyed)	Scorched, completely collapsed, partially/ completely covered with water/mud, or otherwise no longer present.

Image Pairs

Santa Rosa Wildfire
USA 2017

Palu Tsunami
Indonesia 2018

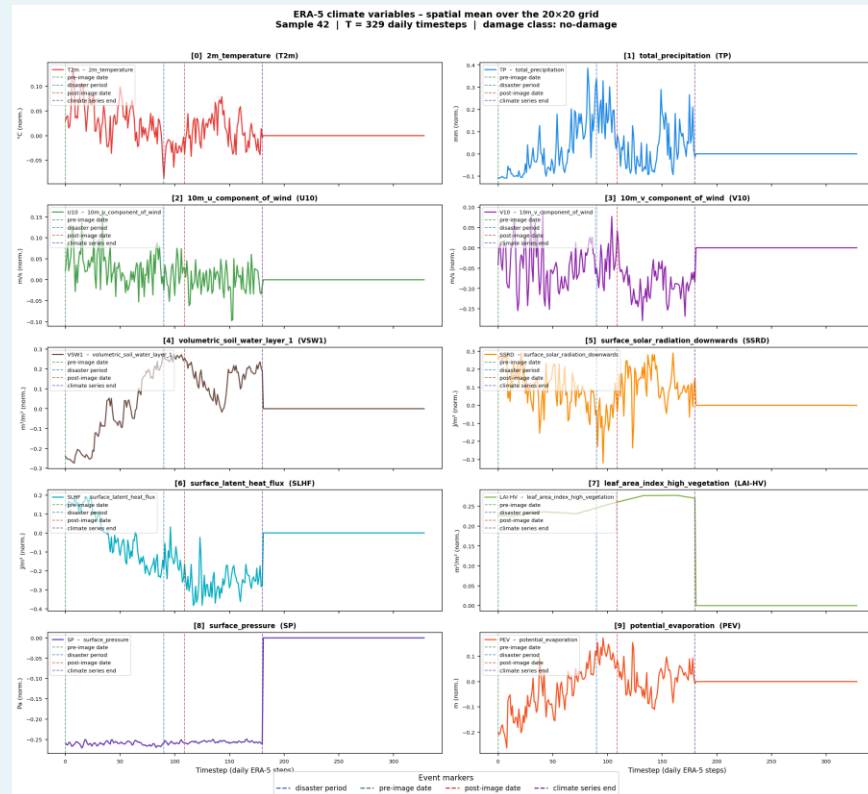


Pre-disaster

Post-disaster

xBD Dataset

Adapted from [Gupta et al., 2019 / xView2]



Multimodal Inputs



Pre-event Image

(3, 64, 64)

RGB satellite patch of the building before the disaster



Post-event Image

(3, 64, 64)

Same building after the disaster — captures visible damage



ERA-5 Climate

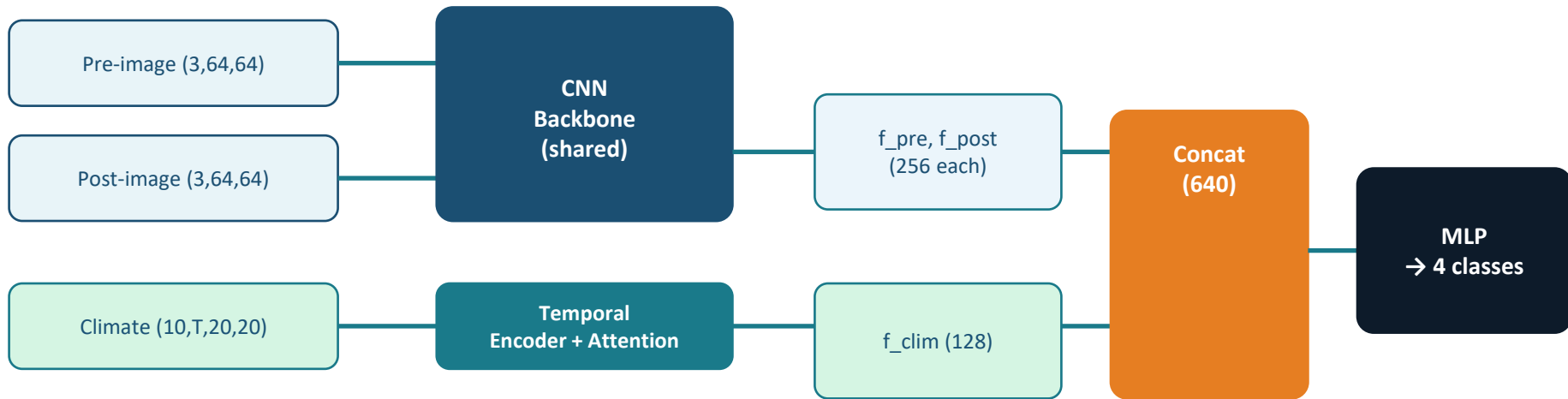
(10, T, 20, 20)

10 atmospheric variables on a 20×20 grid across T daily timesteps

ERA-5 Climate Variables:

- Air Temperature 2m
- Total Precipitation
- U/V Wind Components
- Volumetric Soil Water
- Surface Solar Radiation
- Surface Latent Heat Flux
- Leaf Area Index
- Surface Pressure
- Potential Evaporation

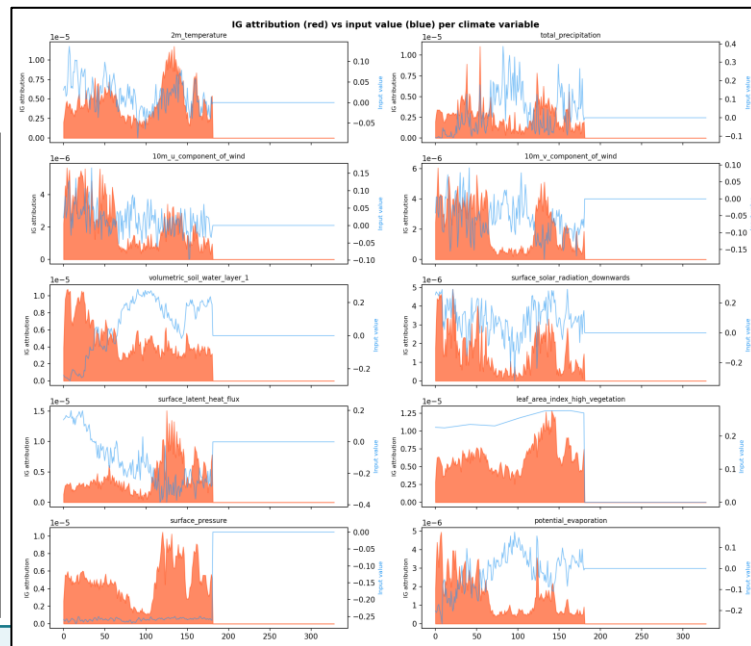
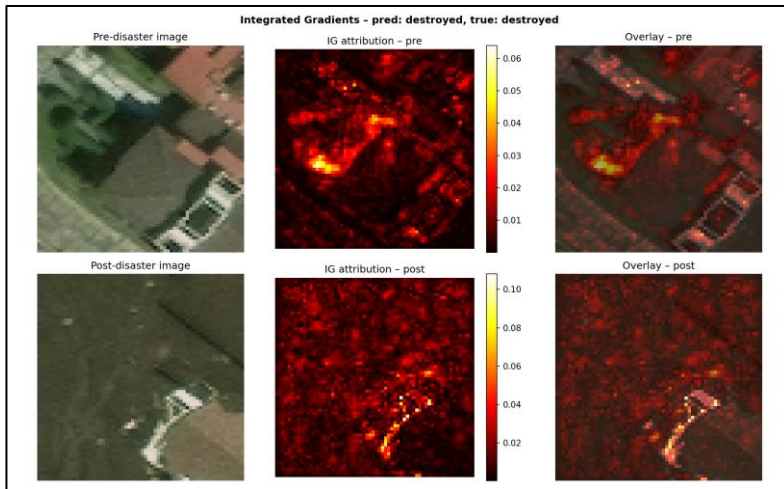
Baseline Model and Setup



Key Design Choices

- ✓ Shared CNN backbone — Same weights for pre- & post- images
- ✓ Attention mechanism over climate timesteps — Learns which days/time steps before/after the event matter (no event-timing labels needed)
- ✓ XAI baseline: Integrated Gradients
- ✓ Focal Loss — Handles heavy class imbalance (buildings mostly no-damage)
- ✓ Evaluation metrics: F1 macro, F1 weighted, confusion matrix

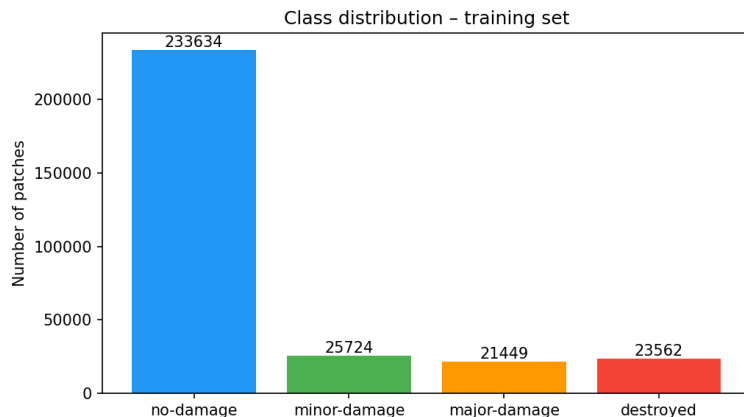
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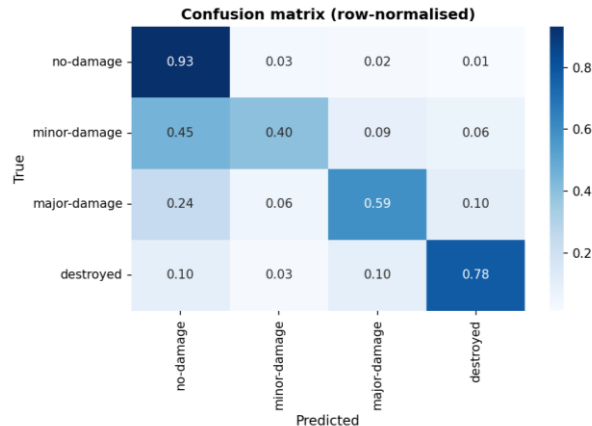
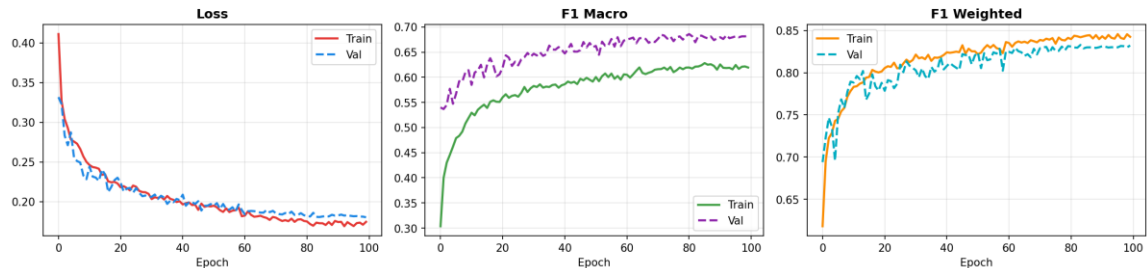


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Baseline Model and Setup

Training curves – BaselineMultimodalModel



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Challenge Deliverables (Wed 17th June, 10:30 AM)

`models/my_model.py`

Your improved model class

`configs/my_config.yaml`

Hyperparameters used

`best.ckpt`

Best checkpoint (highest val F1 macro)

W&B run link

Experiment tracking dashboard

Presentation
(5 slides, 10 minutes maximum)

What you tried and what you learned

Good luck — Build something you're proud of! 🚀